

Leonor Nunes de Carvalho

Lisbon, Portugal
leonor-carvalho.github.io/portfolio/ | leonorncarvalho@tecnico.ulisboa.pt
linkedin.com/in/leonorncarvalho | github.com/leonor-carvalho

Education

Universidade de Lisboa – Instituto Superior Técnico
Master's Degree in Computer Science and Engineering | Current Average: 17.0/20 — Aug. 2024 – Present
Specialisations: *Artificial Intelligence, Bioinformatics & Computational Biology*
MSc Thesis: *Epigenomic Signatures in Renal Cancer: Exploring the Predictive and Regulatory Roles of Understudied Markers* — Expected completion: 2026

Bachelor's Degree in Biomedical Engineering | Final Average: 17.4/20 — Sep. 2021 – Feb. 2024
BSc Final Project (19/20): *A Novel Human-Machine Interface for Surgical Flow Disruption Prevention*

NOVA Medical School – Faculdade de Ciências Médicas
Integrated Master's Degree in Medicine | Current Average: 15.6/20 — Sep. 2024 – Present

About Me

I have always been drawn to interdisciplinary challenges, especially where computational approaches can help make sense of complex biological and clinical questions. My path from Biomedical Engineering to Computer Science has shaped my interest in building technology for healthcare, particularly in biomedical data analysis, clinical decision support and precision medicine. I value understanding not only how a technical solution works, but also the context in which it will be used and the people it is meant to support. That is why I am also studying Medicine: it helps me stay connected to that perspective.

Experience

Computational Biology Teaching Assistant | Instituto Superior Técnico — 2025 & 2026
- Lead lab sessions on sequence alignment, probabilistic models, statistical learning for omics and clinical datasets, and computational functional genomics, supporting MSc students with Python, R and biological interpretation of results.
- Evaluate lab reports, assignments, and exams, and conduct oral assessments with students.

Pre-Clinical Summer Intern | Hospital CUF Tejo – PecliCUF Program — Summer 2026
- Rotated through angiography, cardiology, and the medical-surgical ward, observing stress tests, Holter and ambulatory blood pressure monitoring (ABPM), and learning how these exams are selected, performed, and interpreted in clinical practice.

Research Scholarship Holder | INESC-ID – FRAIL Project — Nov. 2025 – Jan. 2026
- Developed deep learning pipelines on longitudinal health data for knowledge extraction and medical decision support.
- Investigated understudied molecular markers and pharmacological covariates using statistical approaches.

Summer Intern | Hospital da Luz Learning Health — Jul. 2024 – Aug. 2024
- Built a synthetic data pipeline from OMOP-converted clinical records for privacy-preserving healthcare analytics.
- Fine-tuned and evaluated LLMs (GPT-2, Gemma 2) and designed validation frameworks to assess data quality, predictive utility, and privacy preservation.

Leadership & University Involvement

Informatics Coordinator (2025–26) & External Relations Collaborator (2023–24) | NEBM-IST — 2023 – 2026
- Supported company partnerships, networking and internship opportunities, and helped organise two talks with senior industry speakers.
- Coordinated the association's digital infrastructure: website development, technical support, and academic resources.

Biomedical Engineering Summer Camp | Pedagogical Department — Jan. 2025 – Jul. 2025
- Helped launch the first edition for 16 students, coordinating with faculty, delivering a workshop, and supporting participants throughout the event week.

Student Mentor | Programa Mentorado, NAPE-IST — 2021 – 2024
- Mentored 33+ undergraduate, Master's, and mobility students, supporting their integration into academic life at IST.

Jury Member | Hult Prize Local Competition, ISEG — 2026
- Evaluated student startup pitches on feasibility, innovation, and potential impact.

Skills & Additional

Programming & ML: Python, R, MATLAB, Git, Machine Learning, Deep Learning (PyTorch), Statistical Analysis, Data Visualisation
Languages: Portuguese (native), English (fluent), French and Spanish (basic)
Biology: Computational Genomics, Multi-omics, Molecular & Cellular Biology, Physiology
Volunteering & Award: Missão País (since 2022); Banco Alimentar (since 2018); Academic Merit Diploma, IST (2022)